

PART 1: COMPLETE WEBSITE WORKING FLOW — PS 26034

Smart Label Compliance System — Page-by-Page, Click-by-Click

(Ye directly Word/Google Docs me paste karke PDF export karo. Har section PDF ka ek page banega.)

SECTION A: WEBSITE FLOW DOCUMENT

Smart Label Compliance System — Detailed Website Working Flow

Problem Statement ID: 26034

A1. Entry Points & Types

| Portal                   | URL Structure                 | Users                                    |
|--------------------------|-------------------------------|------------------------------------------|
| Main Web Portal          | slcs.gov.in or localhost:3000 | Controller, State Admin, Auditor, Vendor |
| Mobile App               | Android APK                   | Inspector (field work)                   |
| Vendor Self-Audit Portal | slcs.gov.in/vendor            | Manufacturers/Packers/Importers          |

A2. Complete Page Map

|                                              |
|----------------------------------------------|
| LANDING PAGE (/)                             |
| └─ Login Page (/login)                       |
| └─ Forgot Password (/reset)                  |
| └─ OTP Verification                          |
| └─ Role-based redirect after login:          |
| └─ Inspector → [Mobile App]                  |
| └─ District Controller → /dashboard/district |
| └─ State Admin → /dashboard/state            |
| └─ National Admin → /dashboard/national      |
| └─ Vendor → /vendor/dashboard                |
| └─ Auditor → /search (read-only)             |
| └─ Register Pages:                           |
| └─ Vendor Registration (/register/vendor)    |
| └─ Official login (invite, admin creates)    |

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### A3. FLOW 1: AUTHENTICATION & ONBOARDING

Step 1: User opens /login

|

Step 2: Enter email + password

|

Step 3: Backend → POST /api/auth/login

└─ Credentials valid?

| NO → Error: "Invalid credentials" (3 attempts lock)

| YES ↓

└─ JWT access token (15 min) + refresh token (7 days) issued

└─ Role fetched from users table

|

4: FIRST-TIME LOGIN ONLY:

└─ Force password change

└─ 2FA setup (TOTP/SMS OTP)

|

Step 5: Redirect to role-specific dashboard

#### \*\* Registration Flow:\*\*

/register/vendor

└─ Step 1: Company details (name, type: Mfr/Packer/Importer, GST, LUT no.)

└─ 2: Contact + email verification (OTP link)

└─ Step 3: Document upload (registration certificate)

└─ Step 4: Status = "Pending" → District Controller approves

└─ Step 5: Approval email → account active

└─ Validation: GST/LUT format checked live; duplicate company check via registry lookup

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### A4. FLOW 2: INSPECTOR WEB VIEW — Scan Submission & Tracking

*(Primary capture mobile app pe; web pe inspector apne scans review/report karta hai)*

Inspector Login → /inspector/scans

└─ Screen: "My Sc" table

Columns: Date | Product | Shop | Score | Result | Violations | Action

Filters: date range, result, sync status

└─ Click on any scan → /scan/{scan\_id}

└─ Annotated evidence (violations red-boxed)

└─ Extracted fields table with confidence %

└─ Violation list with rule citations

└─ Generated PDF report (View/Download/Re-generate)

└─ Buttons: "Mark Disputed" | "Add Remark"

└─ "New Manual Entry" button (agar app fail ho):

→ Web upload form → drag-drop photos → same AI pipeline → result screen

## A5. FLOW 3: VENDOR SELF-AUDIT PORTAL (Client ka Main Flow)

Vendor Login → /vendor/dashboard

└─ Dashboard shows:

└─ Compliance Score Card (0–100, big number + trend graph)

└─ Total products | Total audits | Open violations count

└─ Recent activity feed

└─ ACTION: "New Self-Audit" → /vendor/audit/new

└─ Step 1: Select/Create product

└─ Existing: search by SKU/barcode from my catalog

└─ New: fill form (name, category, pack size, declared MRP,  
declared net qty, mfg month-year)

└─ Step 2: Upload label artwork

└─ Drag-drop JPG/PNG/PDF (max 20MB each, front/back/sides)

└─ Client-side quality check instantly (blur/resolution warning)

└─ Progress bar → POST /api/audit/upload → job\_id returned

└─ Step 3: Processing screen (live status via WebSocket)

└─ ☐ Image received

|  |  |                                                                       |
|--|--|-----------------------------------------------------------------------|
|  |  | └─ □ Text extraction complete                                         |
|  |  | └─ ● Rule validation running... (animated)                            |
|  |  | └─ Typically 10–15 sec                                                |
|  |  |                                                                       |
|  |  | └─ Step 4: RESULT SCREEN — ye demo ka hero screen hai                 |
|  |  | └─ Overall: Compliance Score + PASS/FAIL banner                       |
|  |  | └─ Left panel: annotated image (green □ boxes on OK zones,            |
|  |  | red X boxes on violations)                                            |
|  |  | └─ Right panel: checklist view                                        |
|  |  | └─ □ MRP present & formatted — Rule 6(1) PASS                         |
|  |  | └─ X Font size 0.7mm < required 1mm — Rule 9(6) FAIL                  |
|  |  | └─ ⚠ Consumer care missing — Rule 6(5) SUSPECTED                      |
|  |  | └─ Each violation row expandable:                                     |
|  |  | └─ detected vs expected value + fix suggestion + rule text            |
|  |  | └─ Bottom buttons:                                                    |
|  |  | └─ [Download Report] [Fix & Resubmit] [Accept Risk*]                  |
|  |  |                                                                       |
|  |  | └─ Loop: Fix & Resubmit → back to Step 2 (version history maintained) |
|  |  |                                                                       |
|  |  | └─ ACTION: "My Products" → /vendor/products                           |
|  |  | └─ Table of SKUs with last audit score/date                           |
|  |  | └─ Compliance History timeline per product (line chart)               |
|  |  |                                                                       |
|  |  | └─ ACTION: "Alerts" → /vendor/alerts                                  |
|  |  | └─ Inspection-based violations on my products (from field scans)      |
|  |  | └─ Each alert → evidence image + report + option to respond/dispute   |
|  |  | └─ Dispute flow: write reason → goes to controller review queue       |
|  |  |                                                                       |
|  |  | └─ ACTION: "Reports" → /vendor/reports                                |
|  |  | └─ All generated PDFs, filterable by product/date/result              |

## A6. FLOW 4: DISTRICT CONTROLLER DASHBOARD

Controller Login → /dashboard/district

└─ TOP KPI CARDS:

- | └─ Today's scans | Violations found | Critical violations
- | └─ Pending notices (approval needed) | Active inspectors
- | └─ Each card clickable → filtered drill-down list

└─ MAP VIEW (Leaflet/Mapbox):

- | └─ District map with violation heat markers (GPS-tagged scans)
- | └─ Click marker → popup: shop name, product, score, top violation
- | └─ Filter by date/severity/product-category

└─ TAB: Notice Approvals → /notices/pending

- | └─ Queue of auto-drafted Section 36 notices (from CRITICAL violations)
- | └─ Review screen draft text (editable) + all evidence attached
- | └─ Actions: [Approve & Issue] [Edit] [Reject with reason]
- | └─ Issued notice → email/SMS to manufacturer + status tracked

└─ TAB: Inspectors → /inspectors

- | └─ Team list: name | today's scans | violations found | avg score
- | └─ Assign priority-inspection items from risk queue
- | └─ Anomaly flags (unusual scan patterns)

└─ TAB: Repeat Offenders → /offenders

- | └─ Brands ≥3 violations, sorted by risk score
- | └─ "Add to Priority Queue" action

└─ TAB: Reports → MIS export (district monthly summary, PDF/XLSX)

## A7. FLOW 5: STATE/NATIONAL ADMIN DASHBOARD

Admin Login → /dashboard/state (ya /national)

- |
- └─ National Heat-map: state-wise violation density (choropleth)
- └─ Trend Charts: monthly compliance %, category-wise violations
- └─ Platform Leaderboard: e-comm platforms ranked by violation rate
- └─ Top Violating Brands table → click → full history
- └─ RULES MANAGEMENT → /admin/rules (☆ differentiator)
  - | └─ Table of codified rules: rule\_no | title | severity | active | version
  - | └─ [Edit Rule] → form with effective\_from date picker
  - | └─ [Add Amendment] → new version created, old retained for history
  - | └─ Audit-logged (who changed what when)
- └─ USER MANAGEMENT → create/deactivate controllers, inspectors, auditors
- └─ AUDIT LOG VIEWER → immutable trail of every system action

## A8. FLOW 6: SEARCH & REPOSITORY (Auditor + All Roles)

- /search
- └─ Search bar: barcode / brand name / SKU / company name
  - └─ Results table: product thumbnail | brand | company | total scans |
    - | latest score | violation count | risk tier
  - └─ Click product → Product Profile Page:
    - | └─ All past scans (timeline)
    - | └─ Compliance history chart
    - | └─ Evidence gallery
    - | └─ Linked notices/alerts
  - └─ Export selected records as CSV/PDF

## A9. BACKEND PROCESSING FLOW (Har Upload Ke Peeche)

POST /api/upload (image + metadata + GPS + JWT)

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└─ Gateway: auth verify → RBAC check → rate limit → audit log
|
└─ Validate: file type (JPG/PNG/PDF), size ≤20MB, EXIF integrity
|
|   (camera-taken? metadata signed? gallery upload blocked for inspectors)
|
└─ Store original → MinIO/S3 → push job to Redis/Celery queue
|
|   → return 202 {job_id} immediately ← user is NOT blocked
|
└─ WORKER PIPELINE (async, parallel where possible):
|
|   1. Preprocess: deskew, denoise, glare removal, quality gate
|
|       ↳ quality < threshold → job fails with "retake photo" reason
|
|   2. Region detection (YOLO): front panel, price zone, back panel,
|
|       sticker regions
|
|   3. OCR (PaddleOCR): multilingual text blocks + positions
|
|   4. NER extraction: structured fields (mfr_name, net_qty, MRP,
|
|       month_year, care_details, origin)
|
|   5. Font measurement: char pixel-height ÷ calibration scale (/mm)
|
|       = physical height → compare Rule 9(6) threshold
|
|   6. Dual-MRP/sticker anomaly CV check
|
|   7. RULE ENGINE: run applicable rules (GPS/state-aware versions)
|
|       → violations[] with rule_no, severity, evidence box coords
|
|   8. Cross-match: QR/barcode registry check + e-comm listing compare
|
|   9. Scoring: weighted score → PASS/PARTIAL/FAIL
|
|   10. Persist: scan_record, extracted_fields, violations → PostgreSQL
|
|       annotated image + PDF report → S3
|
|   11. TRIGGERS: critical?→draft notice | repeat offender?→escalate
|
|       | push notification to submitter
|
└─ Client receives WebSocket event → fetches GET /api/scan/{job_id}

```

## A10. API Endpoint Summary

| Method | Endpoint        | Role | Purpose   |
|--------|-----------------|------|-----------|
| POST   | /api/auth/login | All  | JWT issue |

| Method | Endpoint                     | Role                 | Purpose                  |
|--------|------------------------------|----------------------|--------------------------|
| POST   | /api/audit/upload            | Vendor/Inspector     | Submit label images      |
| GET    | /api/scan/{job_id}           | Owner                | Poll result              |
| WS     | /ws/jobs                     | All                  | Live processing updates  |
| GET    | /api/scans?filters           | Inspector/Controller | List scans               |
| GET    | /api/reports/{id}/pdf        | Authorized           | Download report          |
| POST   | /api/notices/draft           | System               | Auto-draft (internal)    |
| PUT    | /api/notices/{id}/approve    | Controller           | Issue notice             |
| GET    | /api/dashboard/stats         | Role-scoped          | KPI data                 |
| CRUD   | /api/admin/rules             | Admin                | Rules management         |
| GET    | /api/search?q=               | All                  | Repository search        |
| POST   | /api/violations/{id}/dispute | Vendor               | Dispute flow             |
| POST   | /api/batch/ecommm-scan       | System/Admin         | Bulk e-comm scan trigger |